

Multiple Choice (10 marks)

1. _____ is also a part of darknet that is employed for transferring files anonymously.
 - (a) Freenet
 - (b) ARPANET
 - (c) Stuxnet
 - (d) Internet
2. The _____ is categorized as an unknown segment of the Deep Web which has been purposely kept hidden & is inaccessible using standard web browsers.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Dark web
 - (d) Surface web
3. A/an _____ is a program that steals your logins & passwords for instant messaging applications.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) Trojan-Downloader
 - (d) Ransom Trojan
4. What does EXE do when the STP solver times out on a constraint query for a particular path?
 - (a) Assume that the query is satisfiable and continue executing the path.
 - (b) Assume that the query is not satisfiable and stop executing the path
 - (c) Restart STP and retry the query, up to a limited number of retries.
 - (d) Remove a subset of the constraints and retry the query.
5. _____ is the central node of 802.11 wireless operations.
 - (a) WPA
 - (b) Access Point
 - (c) WAP
 - (d) Access Port

True / False (10 marks)

Answer True or False

6. True or False: A session symmetric key between two parties is utilized only once for the duration of that session to ensure confidentiality.
7. True or False: The four Primary Security Principles related to messages include Confidentiality, Access Control, Integrity, and Non-repudiation.
8. True or False: A ping sweep is utilized to identify live systems within a network by sending ICMP echo requests.
9. True or False: A MAC that consistently produces a 5-bit tag is vulnerable to brute-force attacks due to the limited tag space.
10. True or False: WPS, or Wi-Fi Protected Setup, was introduced to simplify the process of connecting new devices to wireless networks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the strengths and weaknesses of various encryption standards in cybersecurity, specifically focusing on why WEP is considered the least secure option.
12. Discuss the concept of buffer-overflow vulnerabilities in software applications, including their causes, potential impacts, and strategies for mitigation in cybersecurity.

End of Paper